$A \times A \times A \times B$,

for any numbers A and B satisfying the following conditions.

1. $R_1 > A$.

2. $R_2 R_3 R_4 > A^2 B$.

Construction:

Let I be a quasi-isometric embedding of $A \times A \times B$ into $R_2 \times R_3 \times R_4$. The conditions above guarantee that such an embedding exists.

Let U be the union of $A \times \text{image(I)}$ with $\{0\} \times R_2 \times R_3 \times R_4$.

Step 1. There is a Lipshitz retraction of R onto U.

Step 2. There is a retraction of U onto $A \times \text{image(I)}$. This retraction has no bound on its 2-dilation, but it does have the following good property. For each point in U, either the retraction is 2-contracting or else that point is mapped into the plane $x_1 = 0$ and its tangent space is mapped into the tangent space of that plane.

Step 3. There is a quasi-isometric map from $A \times \text{image(I)}$ to $A \times A \times A \times B$, which preserves the coordinate x_1.

Step 4. There is a degree 1 Lipshitz pinch map from this rectangle onto itself, whose restriction to the plane $x_1 = 0$ is $p(0, x_2, x_3, x_4) = (0, 0, 0, x_4)$.

This pinching maps collapses all of the region with large 2-dilation to the line $x_1 = x_2 = x_3 = 0$. Since this line is one-dimensional, the 2-dilation of the composition is less than C everywhere.

This pinching map provides the non-linear diffeomorphism that we need to prove Theorem 8.1. Let A be equal to $S_2^{1/2} S_3^{1/2}$, and let B be equal to $S_2^{-1/2} S_3^{1/2} S_4$. There is a 2-contracting linear diffeomorphism from $A \times A \times A \times B$ to S. There is a 2-contracting pinching map from the unit cube to A provided that $1 > CA^2 = CS_2 S_3$, and that $1 > CA^2 B = CS_2^{1/2} S_3^{3/2} S_4$. Since the second inequality implies the first inequality, there is a 2-contracting diffeomorphism from the unit cube to S whenever $S_2 S_3^3 S_4 < c$. This finishes the proof of Theorem 8.1.

5. Double Pinching Map

This degree 1 map takes

$R_1 \times R_2 \times R_3 \times R_4$